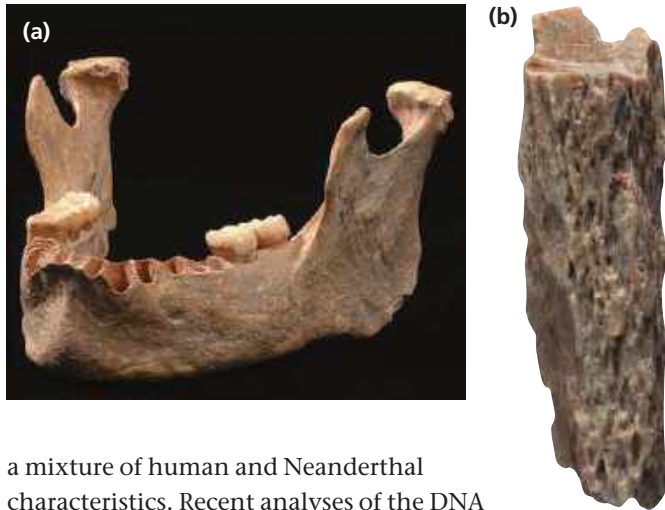


▼ **Figure 34.52 Fossil evidence of interbreeding among hominin species.** (a) This jawbone belonged to a human who lived 40,000 years ago and had a relatively recent Neanderthal ancestor. (b) This bone fragment came from a woman who had a Neanderthal mother and a Denisovan father.



a mixture of human and Neanderthal characteristics. Recent analyses of the DNA sequence of the Neanderthal genome indicated that limited gene flow did occur between the two species. In 2015, the most extensive evidence yet of such gene flow was reported: DNA extracted from the fossil of a human jawbone was found to contain long stretches of Neanderthal DNA (Figure 34.52a). In fact, the amount of Neanderthal DNA in this fossil indicated that this individual's great-great-great-grandparent was a Neanderthal.

Other recent genomic studies have shown that gene flow also occurred between Neanderthals and the Denisovans, whose DNA was isolated from bone fragments discovered in a Siberian cave. In 2018, for example, DNA analyses revealed that the 90,000-year-old bone fragment shown in Figure 34.52b came from a woman who had a Neanderthal mother and a Denisovan father—a remarkable discovery documenting the direct offspring of two different species of hominins. Genomic analyses also indicate that gene flow occurred between the Denisovans and our own species, *H. sapiens*.

Homo sapiens

Evidence from fossils, archaeology, and DNA studies has improved our understanding about how our own species, *Homo sapiens*, emerged and spread around the world.

Fossil evidence indicates that the ancestors of humans originated in Africa. Older species gave rise to later species, ultimately including *H. sapiens*. A 2017 study indicates that by 315,000 years ago, some hominin populations in Morocco had facial features like those of our species, while the back of the skull remained elongated, as in earlier species. The oldest known fossils of our species include

specimens from Ethiopia that are 195,000 and 160,000 years old (Figure 34.53).

These early humans had less pronounced browridges than those found in *H. erectus* and Neanderthals, and they were more slender than other recent hominins. The Ethiopian fossils support inferences about the origin of humans from molecular evidence. DNA analyses show that Europeans and Asians share a relatively recent common ancestor and that many African lineages branched off more basal positions on the human family tree. These findings suggest that all living humans have ancestors that originated as *H. sapiens* in Africa.

The oldest fossils of *H. sapiens* outside Africa are from the Middle East and date back about 180,000 years. Fossil evidence and genetic analyses suggest that humans spread beyond Africa in one or more waves, first into Asia and then to Europe and Australia. The date of the first arrival of humans in the New World is uncertain, although the oldest generally accepted evidence puts that date at about 15,000 years ago.

New findings continually update our understanding of the human evolutionary lineage. For example, in 2015, the human family gained a new member, *Homo naledi*. The structure of its foot indicates that *H. naledi* was fully bipedal, and the shape of its hand suggests that *H. naledi* had fine motor skills (Figure 34.54), as in *H. sapiens*, Neanderthals, and other species that used tools extensively. But *H. naledi* also had a small brain, a broadly flared upper pelvis, and other features that have led researchers to conclude that it was an early member of our genus.

As an early member of our genus, it is likely that *H. naledi* originated more than 1.5 million years ago—yet initial estimates of the age of the *H. naledi* fossils ranged from 3 million

▼ **Figure 34.53 A 160,000-year-old-fossil of *Homo sapiens*.**



▼ **Figure 34.54 Fossils of hand bones and foot bones (top and side views) of *Homo naledi*.**

